

EXAMINATION 1 | CHEM 2313 | FALL 2019

Write your final answers in the boxes provided below. Additional paper has been provided at the end of the exam for scratch work along with supplemental tables. Only pencil, paper, and a model kit are permitted. Write all answers legibly and draw all structures clearly with sensible bond angles.

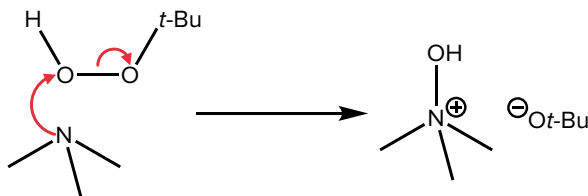
Name

Section	Total Points
I. Multiple-choice Questions	20
II. Predict the Products	20
III. Short-answer Questions	18
III. Acid-base Equilibria	12
IV. Synthetic Planning	10
V. Reaction Mechanism	20

I. Multiple-Choice Questions

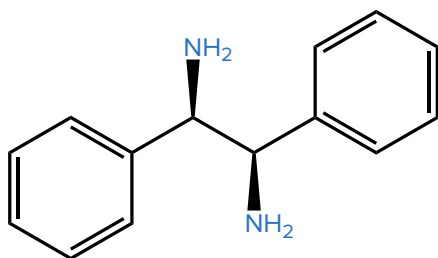
Choose the best answer among the choices provided. Circle your choice *and* write its letter in the box next to each question.

1. What elementary step is depicted in the figure below?



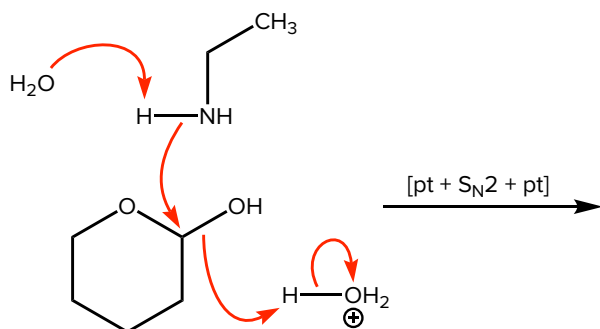
- A. Beta-elimination (E_β)
- B. Bimolecular nucleophilic substitution (S_N2)
- C. Nucleophilic addition (Ad_N)
- D. Proton transfer (pt)
- E. Dissociation of a nucleofuge (D_N)

2. What is the stereotopic relationship between the amino groups in the structure below?



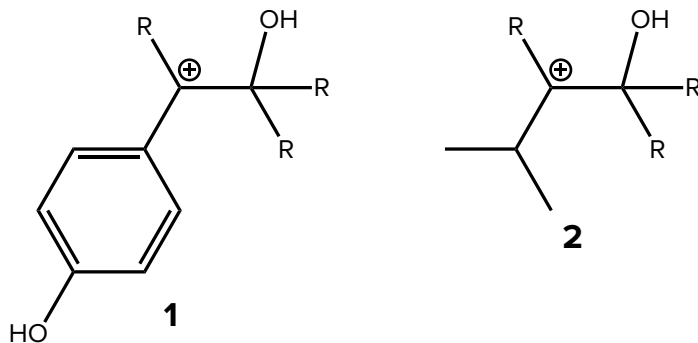
- A. Homotopic
- B. Enantiotopic
- C. Diastereotopic
- D. Constitutionally heterotopic
- E. Additional information is required to answer this question.

3. Which of the following statements is *true* regarding the elementary step below?


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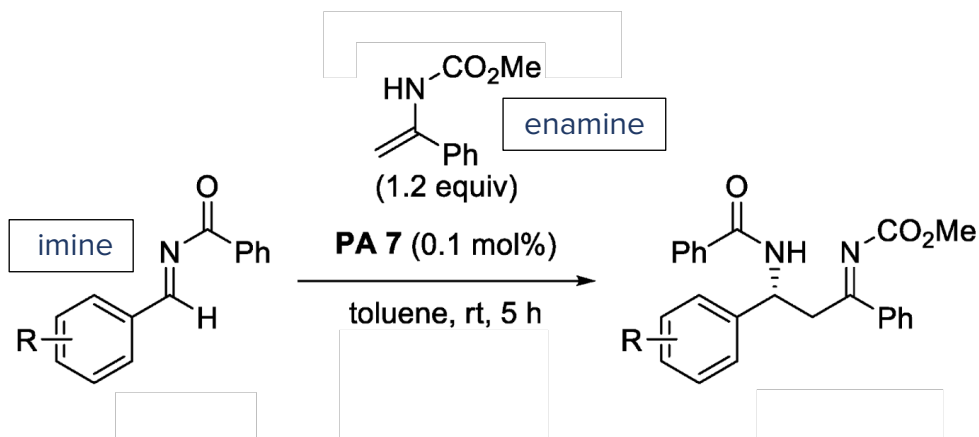
- A. The entropy of activation for this step is large and positive.
- B. The amine is acting as a general acid catalyst.
- C. Water (H_2O) is acting as a general base catalyst.
- D. In the transition state, the amino nitrogen bears substantial positive charge.
- E. The products (not shown) are much higher in energy than the reactants.

4. Which of the following cations is more stable? What stability factor best explains the difference in energy between **1** and **2**?


☐

- A. **1**; steric effect
- B. **1**; resonance delocalization
- C. **1**; inductive effect
- D. **2**; inductive effect
- E. **2**; electronegativity of the charge-bearing atom

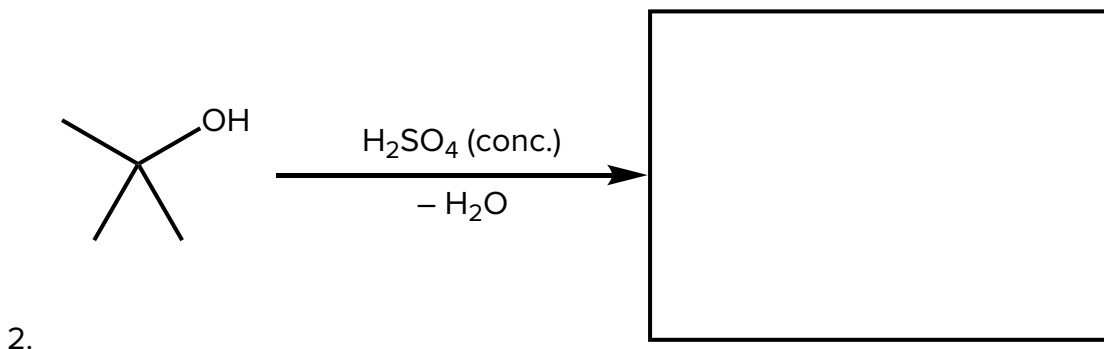
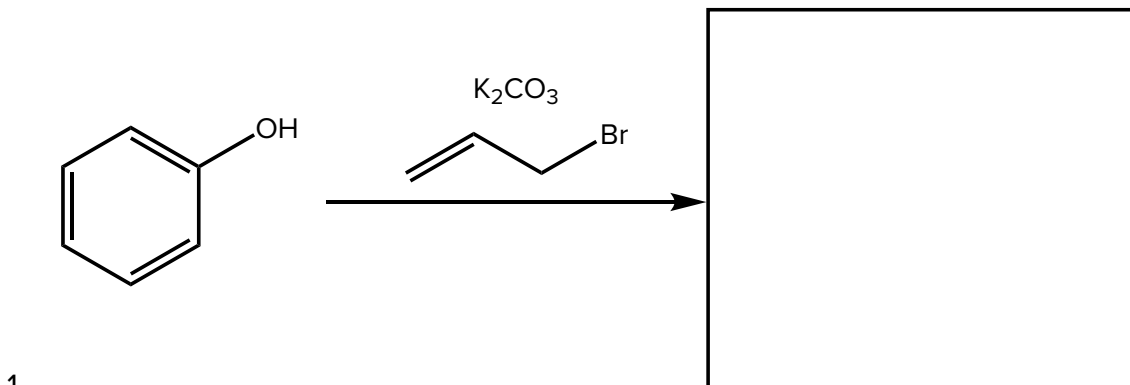
5. Examine the figure below, which illustrates the addition of an enamine nucleophile to an imine. **PA 7** refers to a phosphoric acid catalyst. Which statement best describes the initial action of **PA 7** in this mechanism? The catalyst...

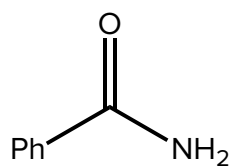

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- A. Protonates nitrogen in the enamine.
 B. Protonates carbon in the enamine.
 C. Deprotonates nitrogen in the enamine.
 D. Protonates nitrogen in the imine.
 E. Coordinates to carbon in the imine.

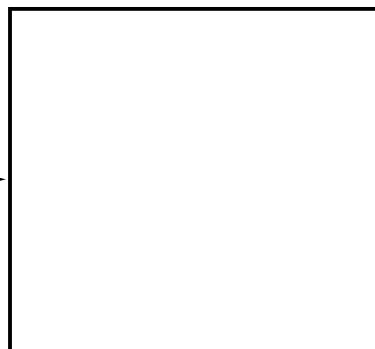
Predict the Products

Provide the major neutral organic product for each reaction or sequence of reactions below after appropriate workup. Clearly indicate the configurations of all stereocenters (except for those with random configurations; stereochemistry can be omitted at these).

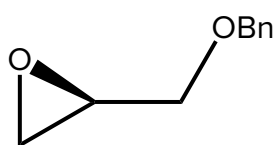




1. Br₂, NaOH
 2. H₂O



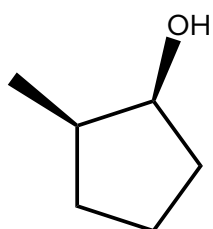
3.



NaH



4.



SOCl₂, NEt₃

dioxane (solvent)

retention of configuration

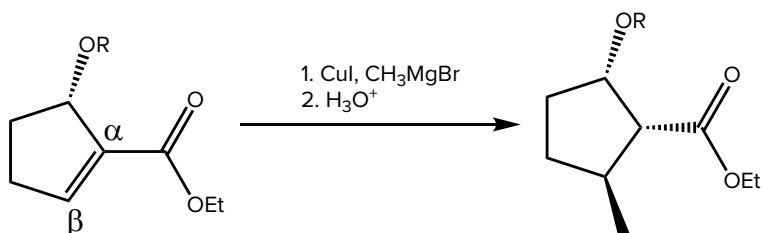


5.

Short-answer Questions

For each question below, provide the requested structure, mechanism, or explanation. Ensure that your complete response fits within the box provided.

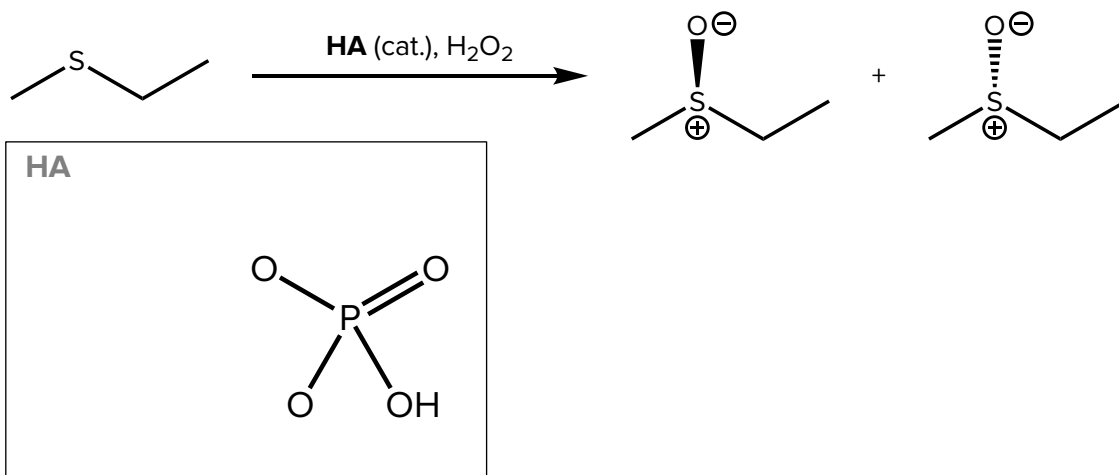
1. The reaction scheme below depicts the addition of a methyl group ("CH₃⁻") to the beta-carbon of an unsaturated ester and addition of hydrogen ("H⁺") to the alpha-carbon. The product shown is the major stereoisomer formed.



Explain why the stereoisomer above forms selectively. Why isn't an equal mixture of the possible stereoisomers observed?



2. The oxidation reaction shown below can be catalyzed by acids. Draw the structure of an acid catalyst **HA** with the general structure $(\text{R}^1\text{O})(\text{R}^2\text{O})\text{P}(=\text{O})\text{OH}$ that would result in the selective formation of one enantiomer of product. A portion of the catalyst structure has been provided. Note that R^1 and R^2 can be identical (or not).

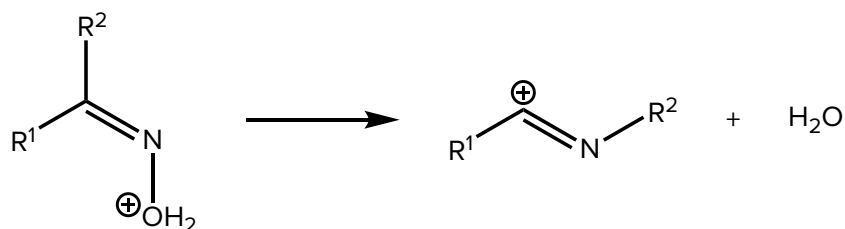


3. In the reaction in the last problem, a thioether reacts with hydrogen peroxide to give a sulfoxide and water (not shown). In the first elementary step in which the thioether and hydrogen peroxide interact, which molecule is the nucleophile and which is the electrophile?

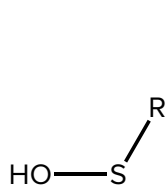
Nucleophile:

Electrophile:

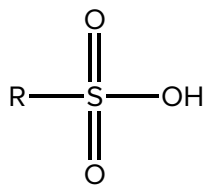
4. The elementary step below is a key step in the Beckmann rearrangement. Draw curved arrows on the reactant structure to show the flow of electrons in this step.



5. Rank the following compounds from most to least acidic by filling in the boxes below.



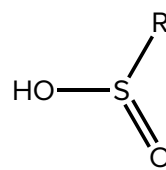
A



B



C



D

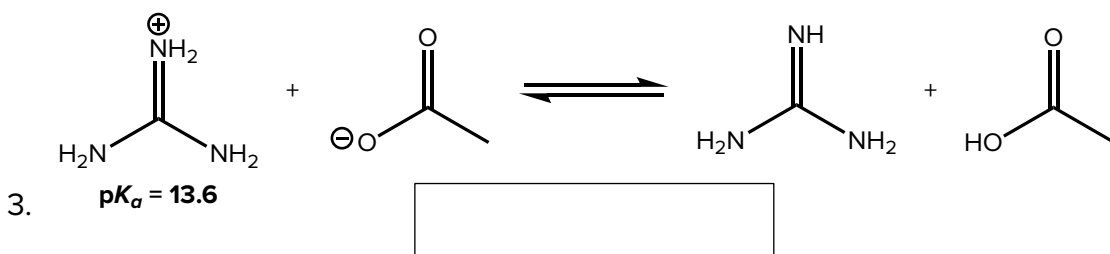
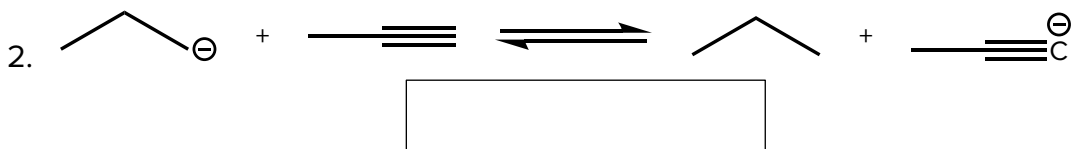
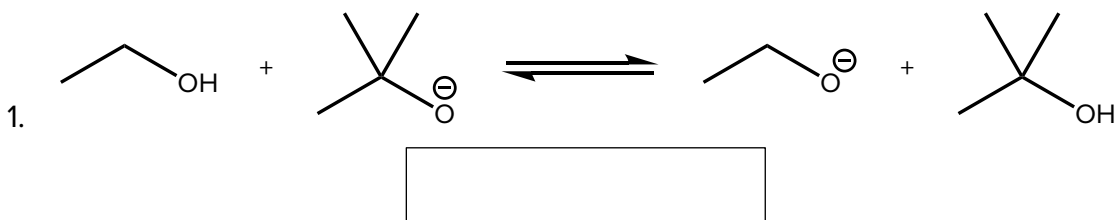


E

Most acidic Least acidic

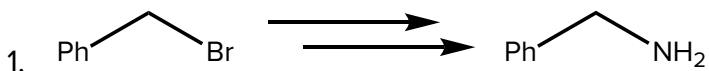
Acid-base Equilibria

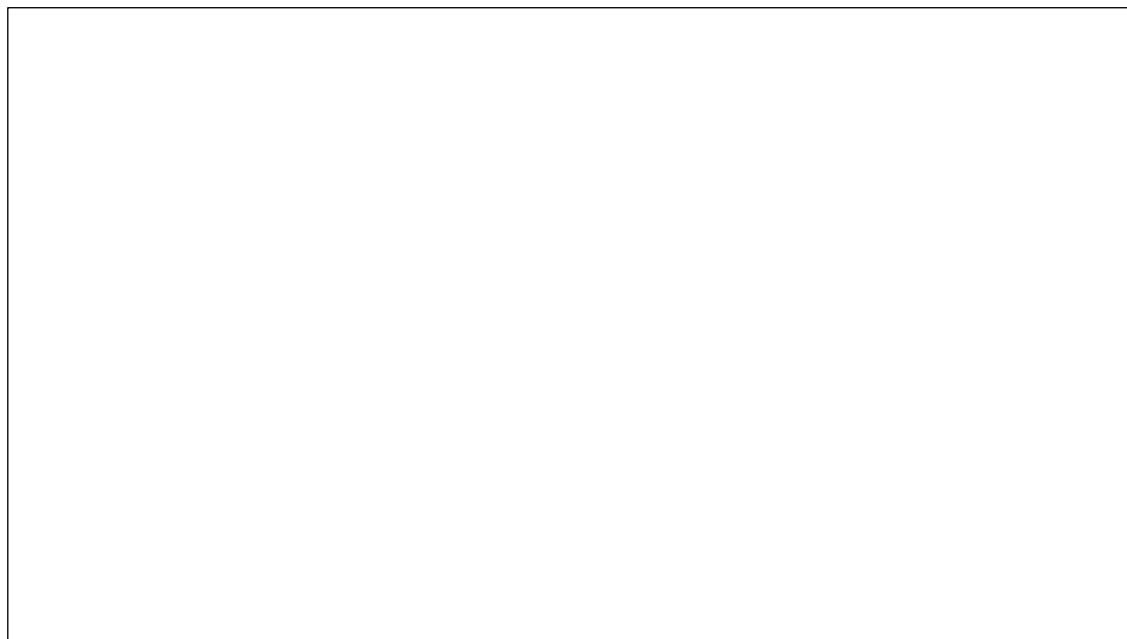
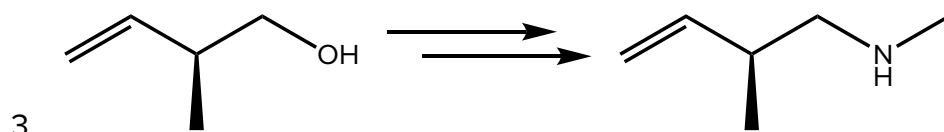
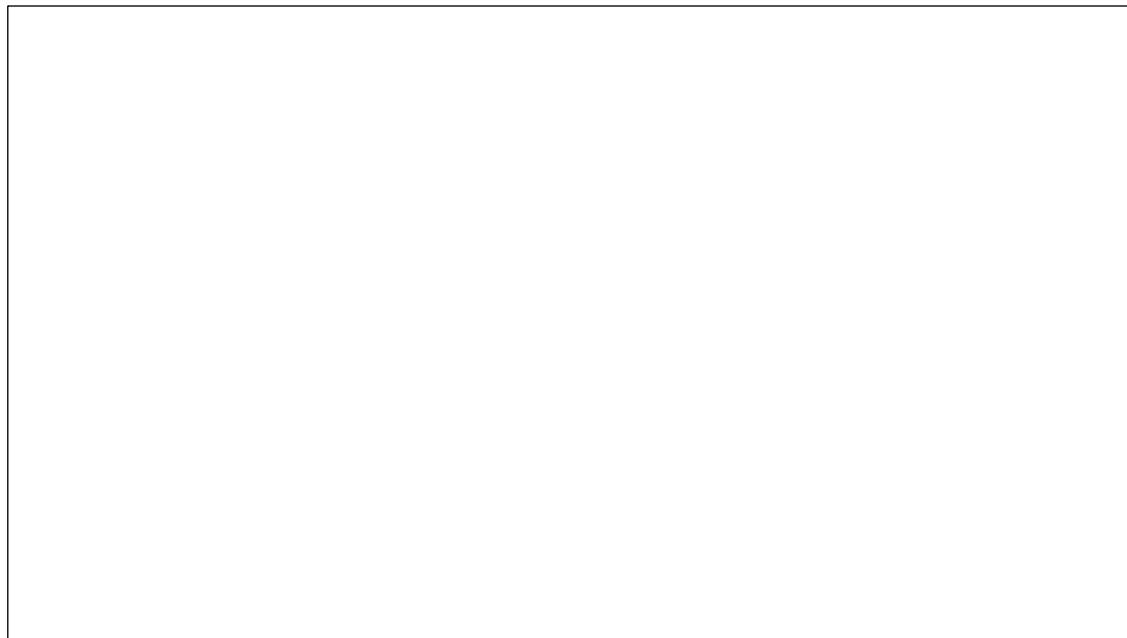
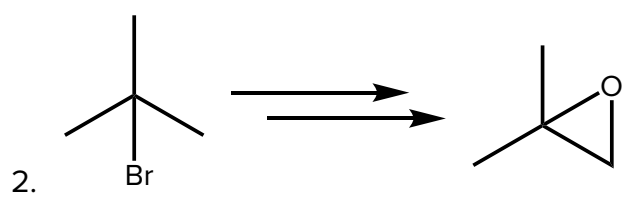
In the box below each reaction, write “products” or “reactants” to indicate which side of the reaction is thermodynamically favored.



Synthetic Planning

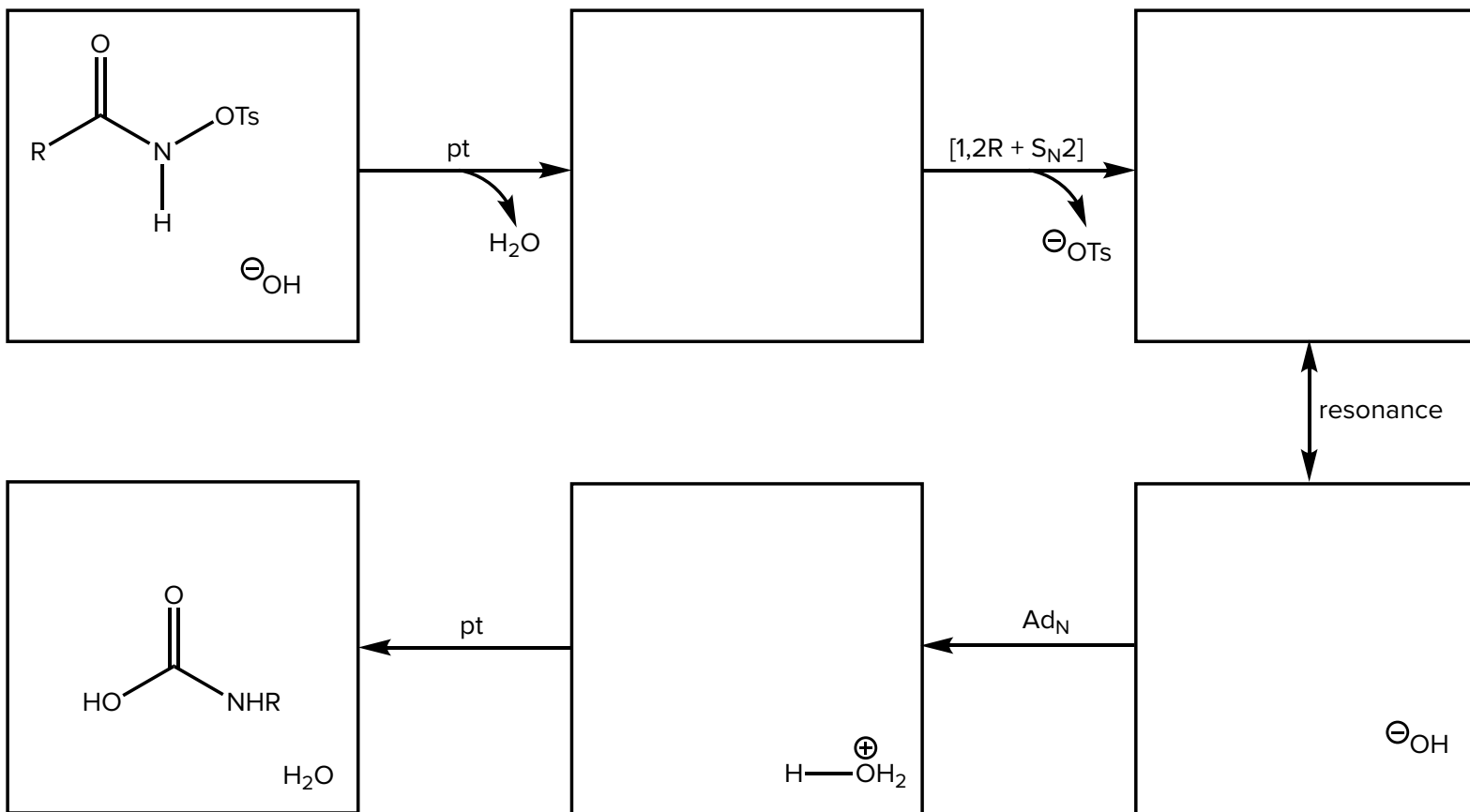
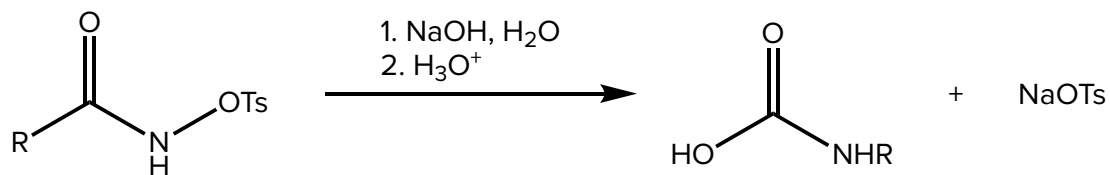
Of the three problems below, **choose two** and provide a sequence of reaction conditions that will accomplish the desired transformation. Include all isolated compounds, but do not draw curved arrows or any reactive intermediates.





Reaction Mechanism

To complete the reaction mechanism below, add curved arrows to the reactants of each elementary step and draw the products in the boxes provided. Each box should include a set of reactants and curved arrows except for the last, which will only include products. Note the step labels provided.



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